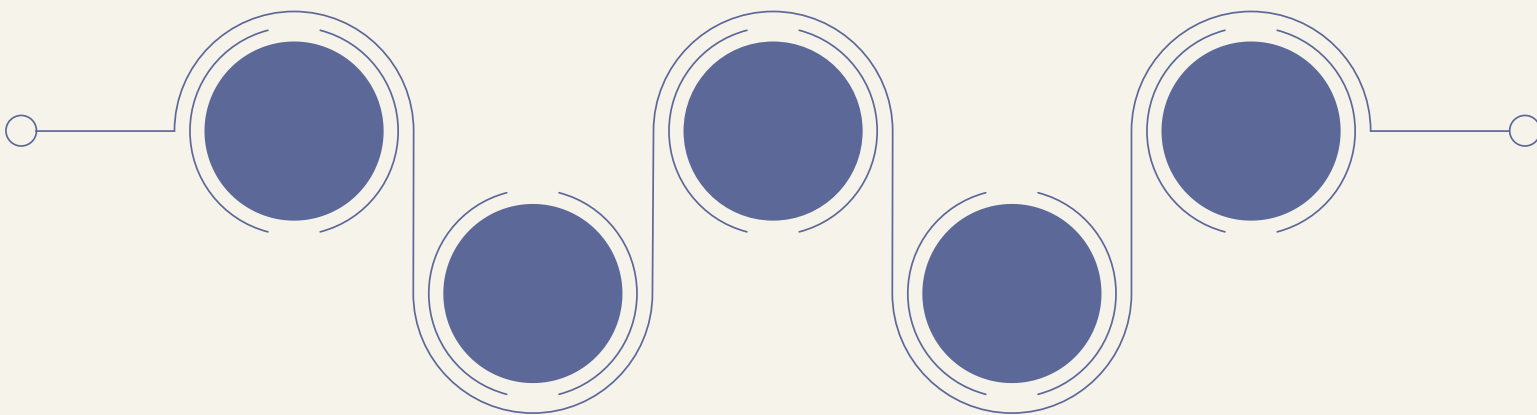


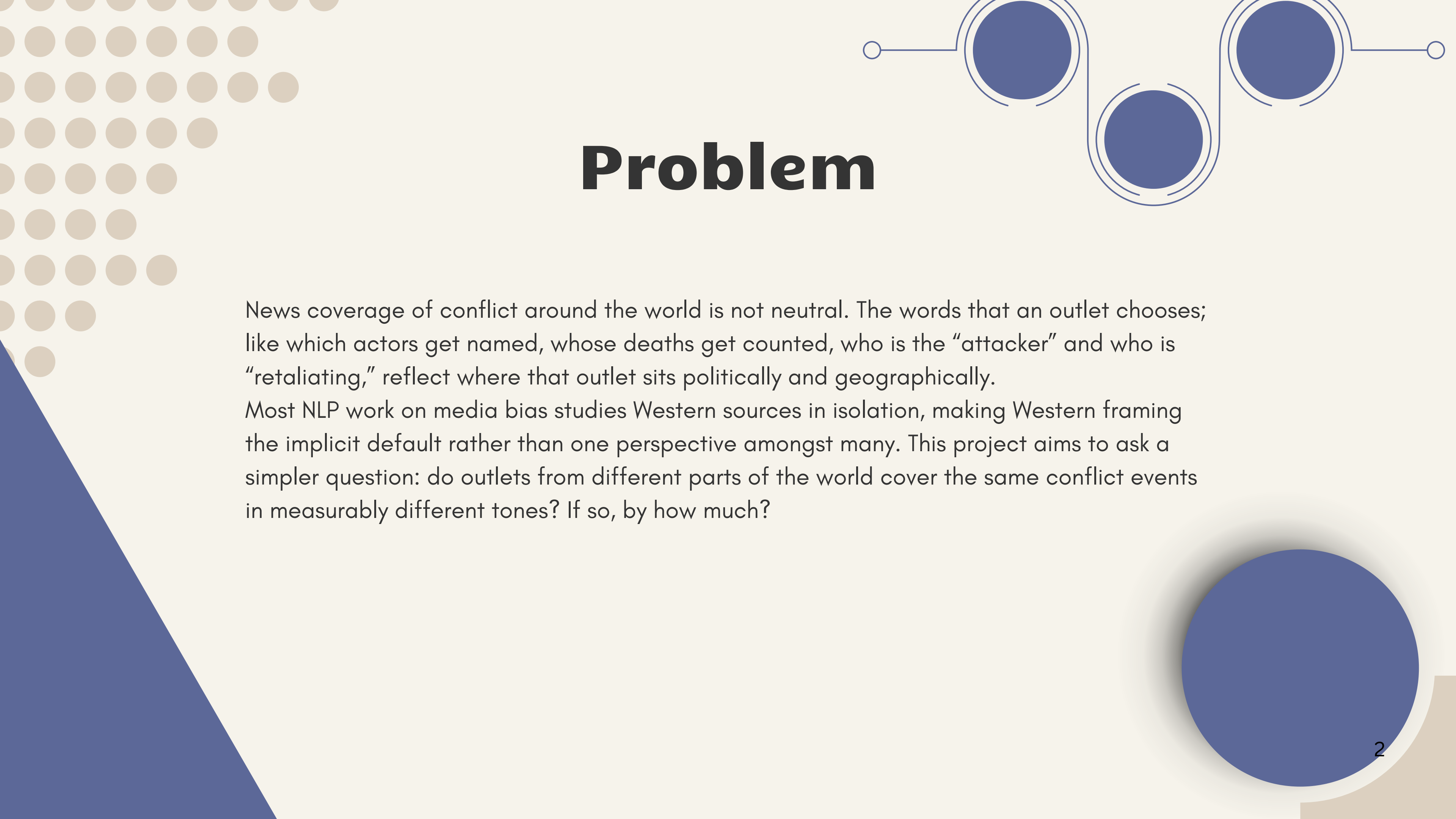


SENTIMENT ANALYSIS OF WESTERN VS. NON- WESTERN CONFLICT NEWS COVERAGE

Presented by Fabian Figueroa

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Problem

News coverage of conflict around the world is not neutral. The words that an outlet chooses; like which actors get named, whose deaths get counted, who is the “attacker” and who is “retaliating,” reflect where that outlet sits politically and geographically.

Most NLP work on media bias studies Western sources in isolation, making Western framing the implicit default rather than one perspective amongst many. This project aims to ask a simpler question: do outlets from different parts of the world cover the same conflict events in measurably different tones? If so, by how much?

Why It Matters

- Framing shapes public understanding of who is a victim, who is an aggressor, and whose suffering counts
- The same event can be seen as a military operation in one outlet and a massacre in another, and not necessarily because the facts differ, but because of word choice and emphasis.
- Computational tools can show these patterns at scale, across hundreds of articles, in ways that manual reading cannot

Datasets and Outlets

- **Western:**

- BBC
- Reuters
- Associated Press
- New York Times

- **Non-Western:**

- Al Jazeera
- RT
- CGTN
- Dawn
- Tehran Times

- **NewsAPI:** full article texts, pulled by outlet name and keyword queries
- **GDELT:** historical coverage archive with built-in tone scores; will be used as a cross-check on model outputs and not as the ground truth



Methods Overview

Method	Type	What it adds
VADER	Lexicon-based	Flat baseline, no labels needed
TextBlob	Lexicon-based	Adds a subjectivity score
Naive Bayes	Supervised	Learned baseline from labelled data
SVM	Supervised	Strong on high-dimensional text
BERT	Transformer	Contextual and handles *hedged language

- **Hedged Language**—Linguistic devices used to indicate uncertainty or modify the commitment to a statement
- **Corpus**—A structured collection of text data used for statistical analysis

Running all five on the same corpus is intentional since where they agree the finding is stronger; where they disagree is itself a result.



Methods: Lexicon & Probabilistic

- **Vader** scores text from -1 to +1 using ~7,500 weighted features. It is fast and requires no labelled data. It will be used as the baseline
- **TextBlob** adds a subjectivity score alongside polarity. A neutral-sounding article can still be opinionated, and this distinction matters when comparing outlet groups
- **Naive Bayes** is a supervised classifier trained on labelled samples. It will establish a learned baseline so model-derived patterns can be compared against what the lexicon tools pick up



Methods: SVM & BERT

- **SVM** can handle high-dimensional text well. Prior work on conflict coverage classification has reached around 91%+ accuracy using bigram features, making it a reasonable benchmark in this context
- **BERT** is fine-tuned on the labelled sample. It has bidirectional attention which handles hedged language and passive constructions that **VADER** and **TextBlob** will miss; i.e. “civilians were reported to have died” reads very different from “forces killed civilians,” but a lexicon tool may score them similarly



Experimental Setup

- **Article Collection: NewsAPI** will be queried by outlet and conflict keyword; filtered to a single event or tight event cluster to hold the topic constant across the comparison
- **Outlet Classification:** will be manually labelled as *Western* or *non-Western*, minimum three outlets per group
- **Labelling: VADER** scores will be used as weak supervisionn across the full corpus; a sample of 200-300 articles manually verified on top of that. **BERT** will be fine-tuned on the verified set



Evaluation Metrics

- **Mean polarity per group:** Western and Non-Western cohorts were compared using a t-test, with bootstrapped confidence intervals computed to quantify effect size uncertainty.
- **Accuracy, Precision, Recall, F1:** for **Naive Bayes**, **SVM**, and **BERT** on the held-out labelled set
- **Spearman Correlation:** The disagreements between **VADER** and **BERT** will be treated as a signal and not noise
- **TextBlob subjectivity distribution:** will be used across outlet groups as a proxy for how opinionated each group's coverage is, separate from its polarity



Challenges & Mitigations

Challenge	Mitigation
Manual Labelling is expensive at scale	VADER weak supervision first, then manual review only on disagreements
RT and CGTN complicate the Western/NonWestern binary	Acknowledge ambiguity; consider adding a state-affiliated flag as a secondary attribute
BERT risks overfitting on a small labelled set	Fall back to an off-the-shelf sentiment checkpoint if the verified sample is too small
GDELT tone scores don't map cleanly to VADER's scale	Treat GDELT as a directional cross-check only, not a ground truth comparison

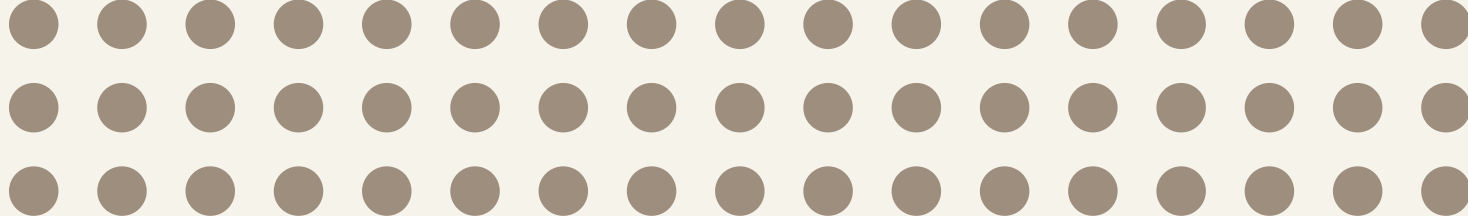


Expected Output

The primary result is a cross-method sentiment comparison: whether Western and non-Western outlets differ in polarity and subjectivity on the same events, and by how much.

Secondary findings:

- Which method is most sensitive to the Western/non-Western distinction
- Where lexicon tools falls short relative to **BERT**
- Whether method agreement correlates with effect size; i.e., are the clearest gaps also the most consistently detected ones



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THANK YOU